This assignment will not be graded and is only for practice.

Level 1

1) Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times3}(\mathbb{R})$) in column A with their bases in column B in Table : M2W4AQ2. **1 point**

	Vector space (Column A)		Basis (Column B)
a)	$V = \left\{ \begin{bmatrix} x & y & 0 \\ 0 & y & 0 \\ 0 & x & x \end{bmatrix} \mid x+y=0, \text{ and } x, y \in \mathbb{R} \right\}$	i)	$\left\{ \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right\}$
b)	$V = \{A \mid A \in M_{3\times3}(\mathbb{R}),\ A \text{ is a diagonal matrix, and the sum of the diagonal entries is zero.} \}$	ii)	$\left\{ \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix} \right\}$
c)	$V = \left\{ \begin{bmatrix} 0 & y & x \\ 0 & 0 & y \\ 0 & 0 & 0 \end{bmatrix} \mid x+y=0, \text{ and } x, y \in \mathbb{R} \right\}$	iii)	$\left\{ \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\}$

Table : M2W4AQ2

Choose the correct option.

- ☐ a → ii
- ☐ a → i
- ☐ b → iii
- ☐ c → i
- ☐ b → ii
- ☐ c → iii

No, the answer is incorrect.

Score: 0

Accepted Answers:

a → ii
b → iii
c → i

2) If S be a subset of $\mathbb{R}^5$ such that $span(S) = \mathbb{R}^5$, then what is the minimum number of elements possible in S?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

3) If S_1 is a maximal linear independent set and S_2 is a minimal spanning set of a vector space V, then which of the following option(s) is (are) true? ***1 point***

- [ ] For any $v \in S_1, S_1 \setminus \{v\}$ is a linearly independent.
- [ ] For any $v \in S_2, S_2 \setminus \{v\}$ is a spanning set of V.
- [ ] For any $v \in V \setminus S_1, S_1 \cup \{v\}$ is a linearly dependent.
- [ ] For any $v \in V, S_2 \cup \{v\}$ is a spanning set of V.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For any $v \in S_1, S_1 \setminus \{v\}$ is a linearly independent.
For any $v \in V \setminus S_1, S_1 \cup \{v\}$ is a linearly dependent.
For any $v \in V, S_2 \cup \{v\}$ is a spanning set of V.

4) Consider the following subset of $\mathbb{R}^2$ with usual addition and scalar multiplication as in $\mathbb{R}^2$. ***1 point***

$V_1 = \{(x, 0) \mid x \in \mathbb{R}\}$
$V_2 = \{(0, y) \mid y \in \mathbb{R}\}$
Which of the following options are correct?

- [ ] Both V_1 and V_2 are subspaces of $\mathbb{R}^2$.
- [ ] $V_1 \cap V_2$ is a subspace of $\mathbb{R}^2$.
- [ ] $V_1 \cup V_2$ is a subspace of $\mathbb{R}^2$.
- [ ] V_1 is a subspace of $\mathbb{R}^2$, but V_2 is not.
- [ ] V_2 is a subspace of $\mathbb{R}^2$, but V_1 is not.
- [ ] $\{(1, 0)\}$ is a basis of V_1.
- [ ] $\{(0, 1)\}$ is a basis of V_2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both V_1 and V_2 are subspaces of $\mathbb{R}^2$.
$V_1 \cap V_2$ is a subspace of $\mathbb{R}^2$.
$\{(1, 0)\}$ is a basis of V_1.
$\{(0, 1)\}$ is a basis of V_2.

5) Consider the following subset of $\mathbb{R}^2$ with usual addition and scalar multiplication as in $\mathbb{R}^2$. ***1 point***
$V_1 = \{(x, 0) \mid x \in \mathbb{R}\}$
$V_2 = \{(2x, 0) \mid x \in \mathbb{R}\}$
Which of the following options are correct?

- [ ] Both V_1 and V_2 are subspaces of $\mathbb{R}^2$.
- [ ] $V_1 \cap V_2$ is a subspace of $\mathbb{R}^2$.
- [ ] $V_1 \cup V_2$ is a subspace of $\mathbb{R}^2$.
- [ ] V_1 is a subspace of $\mathbb{R}^2$, but V_2 is not.
- [ ] V_2 is a subspace of $\mathbb{R}^2$, but V_1 is not.
- [ ] $\{(1, 0)\}$ is a basis of V_1.

- [ ] $\{(1, 0)\}$ is a basis of V_2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both V_1 and V_2 are subspaces of $\mathbb{R}^2$.
$V_1 \cap V_2$ is a subspace of $\mathbb{R}^2$.
$V_1 \cup V_2$ is a subspace of $\mathbb{R}^2$.
$\{(1, 0)\}$ is a basis of V_1.
$\{(1, 0)\}$ is a basis of V_2.

6) Let V be a vector space which is defined as follows: **1 point**

$$V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq \mathbb{R}^4$$

with usual addition and scalar multiplication. Which of the following set forms a basis of V?

- [ ] $\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)\}$.
- [ ] $\{(1, 1, 0, 0), (0, 1, -1, 0), (0, -1, 0, 1)\}$.
- [ ] $\{(1, 0, -1, 0), (1, 1, 0, 0), (1, 0, 0, 1)\}$.
- [ ] $\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0, -1, 0), (1, 1, 0, 0), (1, 0, 0, 1)\}$.
$\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$.

Level 2

7) Which of the following sets form a basis of the vector space of 2×2 lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct) **1 point**

- [ ]

$\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}\}$

- [ ] $\{\begin{bmatrix}1 & 0\\1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\}$
- [ ] $\{\begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}\}$
- [ ] $\{\begin{bmatrix}-1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & -1\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}\}$
$\{\begin{bmatrix}1 & 0\\1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\}$
$\{\begin{bmatrix}-1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & -1\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}\}$

8) If S is a linearly independent set in vector space $V = \{(x, y) \mid y = 2x, \text{ where } x, y \in \mathbb{R}\}$ with usual addition and scalar multiplication as in $\mathbb{R}^2$, then find the maximum possible cardinality of S.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

9) If $\{v_1, v_2, v_3\}$ forms a basis of $\mathbb{R}^3$, then which of the following are true? ***1 point***

- [ ] $\{v_1, v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$
- [ ] $\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$.
- [ ] $\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.
- [ ] $\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$
$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$.
$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.
$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.

10) Which of the following options is(are) true? ***1 point***

- [ ] Any minimal spanning set of a vector space V must be a basis of V.
- [ ] Any maximal spanning set of a vector space V must be a basis of V.
- [ ] Any minimal linear independent set of vector space V must be a basis of V.
- [ ] Any maximal linear independent set of vector space V must be a basis of V.
- [ ] The basis of a vector space is unique.
- [ ] The number of elements in a basis of $\mathbb{R}^3$ is 3.
- [ ] The number of elements in a basis of $M_{3\times3}(\mathbb{R})$ is 3.
- [ ] There are infinite number of bases of $\mathbb{R}^3$.
- [ ] Any subset of a minimal spanning set of V cannot be a spanning set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Any minimal spanning set of a vector space V must be a basis of V.
Any maximal linear independent set of vector space V must be a basis of V.
The number of elements in a basis of $\mathbb{R}^3$ is 3.
There are infinite number of bases of $\mathbb{R}^3$.
Any subset of a minimal spanning set of V cannot be a spanning set.

Your score is: 0/10